

Attention Expanded Note-Taking Framework

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1 Summary

This week we looked at expanding upon attention. Attention based transformers face one major challenge, they are limited to a smaller number of tokens because of computational and memory requirements that are quadratic with the input sequence length. So, several methods have been experimented with to adapt attention to handle larger contexts. These methods include things such as sparse attention, memory augmentation approaches, and making changes to positional embeddings.

Sparse attention methods aim to reduce computational costs by not doing all of the pairwise comparisons between tokens. An example of this is the Big-Bird method, in this method it combines global tokens, local attention, and random attention to perform sparse attention instead of full attention (Constructing Transformers For Longer Sequences with Sparse Attention Methods, Dubey). On the other hand, memory augmentation methods leave the attention methods alone, but introduce either external storage or a retrieval system that allows the model to access prior tokens without keeping them in main/active memory. This creates a sort of "long-term memory" to allow the model to recall older context more efficiently (The What, Why, and How of Context Length Extension Techniques in Large Language Models – A Detailed Survey, Section: Augmenting Language Models with Long-Term Memory). Then there is the approach of editing how the positional embeddings work. The traditional method of encoding the absolute positions fail when going beyond the lengths seen in model training. Techniques to combat this include relative positional embeddings instead of absolute, and rotary position embeddings (RoPE) in which you rotate the query and key at an angle to reflect the absolute position of the embedding in the input sequence (Beyond the Limits: A Survey of Techniques to Extend the Context Length in Large Language Models, Section: Positional Extrapolation and Interpolation).

This is just a few of the methods that have been experimented with when it comes to expanding the abilities of attention based transformers and attention itself. The driving force of all of these experiments is to find the best ways to increase the total capacity of attention based methods so they can reason more effectively on larger/longer contexts.

2 Attention Grabbers

1. I like the visual representation of the self-attention mechanism as a directed graph. It showed how the BigBird process adds global, local, and random connections to construct a directed graph that looks very similar to doing full-attention (complete graph).
2. I think the concept of LLMs "forgetting" long-term context is interesting. It shows that even though they can hold human like conversation, they don't have human like memory. They can't compute similarities as text gets too long.
3. The concept of positional encoding being a large cause for transformers not being able to handle super long context stood out to me. It stood out because I think it makes the most sense, tracking the position of something further away would be more difficult.
4. The section labeled SoftMax-Free attention in the last paper caught my eye immediately. Because I was like that's kind of an important step in the process.
5. The attention-free transformer section also caught my attention right away. Attention is literally the backbone of transformers, so I'm shocked that people would try to make it work without attention. That wouldn't be my first thought of what to change at least.

3 Questions

1. What about something like selective forgetting? Being able to identify parts of a context that isn't important and ignoring it, then we aren't just adding more context just for the sake of it.
2. Is the loss of some of the more fine-grained understanding with full-attention worth being able to handle larger contexts with sparse attention?
3. Is there simply a limit to transformers and we need to look to other architectures for handling longer context?
4. What's the largest benefit to being able to handle longer contexts?
5. Copilot has conversation limits, is this because the models can't remember things from earlier in the chat once the chat gets too long?